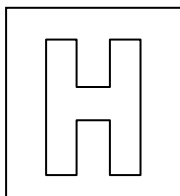


Candidate Name: _____
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Class Adm No

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2021 Prelim Examination Pre-university 3

H2 BIOLOGY

9744/04

Paper 4 Practical

30 August 2021

Candidates answer on the Question Paper

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams and graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer all questions in the spaces provided on the Question Paper.

The use of scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/33
2	/8
3	/14
Total	/55

This question paper consists of 20 printed pages, including 1 blank page.

[Turn over

Candidates with access to microscope at the start of the paper are given the **first 1 hour** to use it. Please answer **QUESTION 3** within this time frame.

Candidates with no access to microscope at the start of the paper will be given access 1 hour 30 mins after the start of the paper. You may proceed with **QUESTION 1 and 2** first.

Answer **all** questions

Question 1

An agar block stained blue with DCPIP gradually becomes colourless when immersed in a solution of ascorbic acid. This is because the colour of DCPIP changes from blue to colourless when it reacts with ascorbic acid.

The end-point is when the agar block is completely colourless.

The concentration of an ascorbic acid solution can be estimated by comparing the time taken to reach the end-point with the times taken for ascorbic acid solutions of known concentrations.

You are provided with:

- 100 cm³ of 0.10% ascorbic acid solution, labelled **V**.
- 100 cm³ of distilled water, labelled **W**.
- Large rectangular block of agar, stained blue with DCPIP.

You are required to find the time taken for different concentrations of ascorbic acid to decolourise pieces of agar stained blue with DCPIP.

To avoid staining your skin, try not to touch the agar. It is recommended that you wear gloves.

Read through steps 1-11 and prepare **(a)(i)**, **(a)(ii)** and **(a)(iii)** before starting the experiment.

Proceed as follows.

1. In the small containers provided, prepare a range of concentrations of ascorbic acid using **V**. Using simple dilution, each successive concentration should be reduced by 0.02%. A control should also be set up.

(a)

- (i)** State the volume of each concentration required to complete the experiment.

Volume of each concentration neededcm³
[1]

- (ii)** Draw a table below to record how you will make the different concentrations of ascorbic acid.

[3]

You may use blunt forceps and the paper towels to handle the agar.

- Place the large rectangular block of agar onto a white tile and cut this block into identical pieces of 5 mm x 5 mm, as shown in Fig. 1.1. The thickness of the block should not be changed.

You will need to consider the number of pieces needed to complete your investigation.

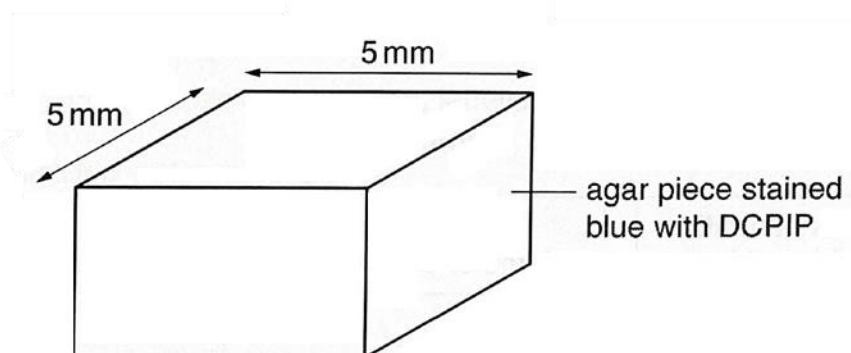


Fig. 1.1

- Label another empty container as **trial**.
 - Put three of the prepared pieces of agar into the 0.10% solution of ascorbic acid in the container prepared in step 1. Ensure that the agar pieces do not stick together.
 - Immediately start timing.
 - Gently swirl the contents of the container at regular intervals.
 - Observe the changes to the pieces of agar to identify the end-point and note the time at which this is reached. This is to help you identify the end-point in your investigation. It may be helpful to view the agar pieces against a white background. It is not necessary to record this result as you will repeat in step 9.
 - Remove these pieces of agar and put them aside in a container labelled **trial**. Keep the 0.10% solution of ascorbic acid for re-use in step 9.
 - Repeat steps 4-6 using the 0.10% ascorbic acid retained from step 8 and the next concentration of ascorbic acid prepared in step 1. These should be set up at the same time.
 - Record, in the table prepared in **(a)(iii)**, the time taken for all three pieces of agar, for each of the ascorbic acid concentrations tested, to reach the end-point you identified in step 7.
- After five minutes, if any piece of agar has not decolourised, stop timing and record 'more than 300' for the concentration of ascorbic acid.
- Repeat step 9 and 10 to obtain results for the remaining concentrations of ascorbic acid and the control.

- (iii) Record your results in a suitable table in the space below.

[3]

12. You are provided with sample **X**. This is a solution of ascorbic acid of unknown concentration.

- (iv) Using suitable volume of sample **X**, repeat the same steps to deduce the time taken to decolourise agar pieces.

Time taken to reach the end-point [1]

- (v) Using your results from **(a)(iii)** and **(a)(iv)**, estimate the concentration of ascorbic acid in sample **X**.

Percentage concentration of ascorbic acid in sample **X**.....
[1]

(b)

- (i)** Explain how a decrease in the size of a piece of agar would affect the time taken to reach the end-point.

.....

.....

.....

..... [2]

- (ii)** Suggest two modifications to the investigation that would improve the accuracy of your estimate in **(a)(v)**.

.....

.....

.....

..... [2]

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- (c) A student used a different indicator to measure the concentration of ascorbic acid in different solutions.

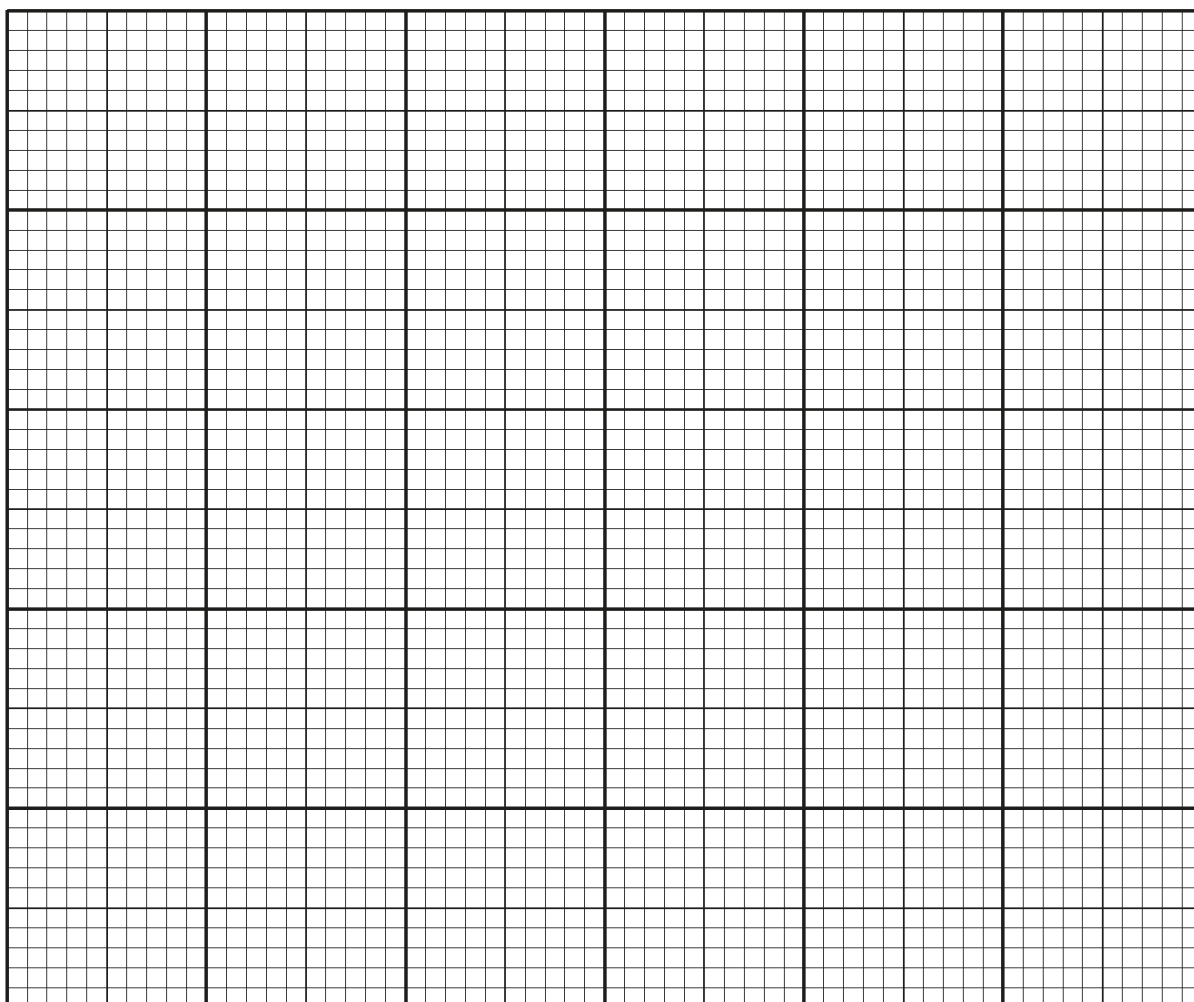
The higher the concentration of ascorbic acid in a solution, the greater the volume of this new indicator required to reach the end-point.

The results for the ascorbic acid solutions of known concentrations are shown in Table 1.1.

Table 1.1

Percentage concentration of ascorbic acid	Volume of indicator (cm ³)
0.24	0.60
0.42	1.05
0.60	1.50
0.78	1.95
0.90	2.25

- (i) Plot the results in Table 1.1. Draw a line of best fit.



[3]

- (i) As the relationship shown in the graph is linear, the gradient of the line can be used to estimate the concentration of ascorbic acid for any measured volume of indicator. This means that it is possible to estimate the concentration of ascorbic acid for values that are greater than those plotted on the graph.

Calculate the gradient of the line.

$$\text{Gradient of the line} = \frac{\text{Change in volume of indicator}}{\text{Change in concentration of ascorbic acid}}$$

Gradient of the line.....[1]

- (ii) Use the gradient from (i) to estimate the concentration of ascorbic acid in a solution requiring 3.50 cm³ of indicator to reach the end-point.

Show your calculation clearly.

Percentage concentration of ascorbic acid[1]

- (d) A student wanted to use DCPIP to determine if reactions in the light-dependent stage had occurred in a suspension of chloroplasts. When reduced by hydrogen ions and electrons, DCPIP turns from blue to colourless.

The student wanted to ensure that either the DCPIP or chloroplast suspension does not result in colour change on its own. The student deduced that 4 different tubes must be set up. One of the tube contained chloroplast suspension and DCPIP and was exposed to light source from the lamp.

The set-up for the experiment is shown in Fig. 1.2.

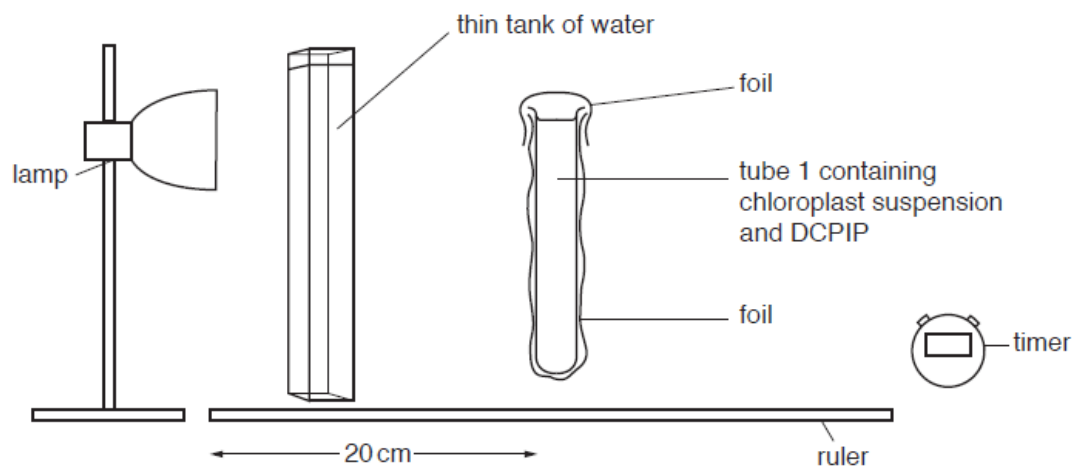


Fig. 1.2

- (i) Complete Table 1.2 by identifying the contents in the 4 tubes. Tube 1 has been completed for you.

Table 1.2

Tube	Contents	Exposed to light
1	Chloroplast suspension and DCPIP	Yes
2		
3		
4		

[3]

- (ii) Suggest the function of the thin tank of water.

.....

..... [1]

- (iii) Suggest why all the tubes should be covered in foil at the start of the experiment.

.....

..... [1]

- (iv) Design an experiment to investigate if light-dependent stage of photosynthesis has occurred in a suspension of chloroplast.

In your plan, you must use:

- Set-up in Fig. 1.2
- Leaf
- Mortar and pestle
- Filter funnel and filter paper
- Centrifuge
- Extraction medium
- DCPIP

You may select from the following sterilized apparatus and plan to use appropriate additional apparatus:

- Normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.
- Syringes

Your plan should:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- Identify the independent and dependent variables
- Describe the method with scientific reasoning used to describe the method so that the results are as accurate and repeatable as possible
- Explain the rationale for control setups
- Include layout of results tables
- Use the correct technical and scientific terms
- Include reference to safety measures to minimize any risks associated with the proposed experiment.

[Turn over

Question 2

You are required to investigate the rate of oxygen uptake by respiring seeds. A respirometer shown in Fig. 2.1 will be used to measure the rate of oxygen uptake.

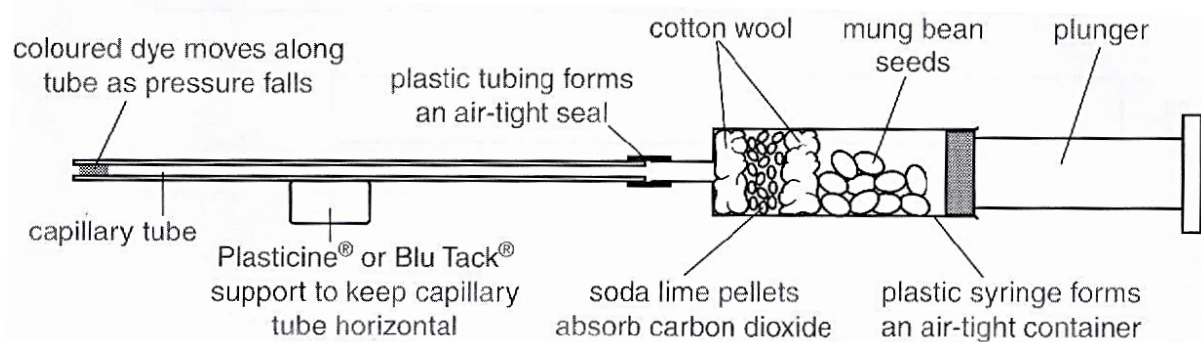


Fig. 2.1

You are provided with:

- Mung bean seeds
- Soda lime

Soda lime is harmful and corrosive. Safety glasses and gloves should be worn.

Read through steps 1-10 and prepare **(a)** before starting the experiment.

Proceed as follows.

1. Place a small plug of cotton wool in the respirometer and use a glass rod to gently push it to the bottom. The cotton wool must NOT be compacted.
2. Using a spatula, add soda lime pellets on top of the cotton wool plug in the respirometer. The soda lime pellets should form a layer of about 1 cm deep.
3. Add another small plug of cotton wool to the respirometer and gently push it down until it is just above the soda lime pellets. Do NOT compact the cotton wool.
4. Take 10 mung beans and briefly dab them dry with paper towels.
5. Add the 10 mung bean seeds to the respirometer. Place the plunger. Leave the respirometer for 5 minutes.
6. Read through the remaining steps up to step 10 and decide on the results that you will be recording.
7. Place the end of the capillary tube into the dye solution and use the syringe plunger to carefully pull in about 1 cm length of the dye. Wipe off any excess dye on the outside of the capillary tube.

8. Place a ruler alongside the capillary tube and use plasticine as support to keep the capillary tube horizontal and aligned with the ruler.
9. Record the **cumulative distance in mm** moved by the dye at **every 30-second intervals** for **5 minutes**. Do not start the stopwatch until the dye has started to move.
10. If the dye has reached the end of the capillary tube before 5 minutes, pause the stopwatch but do not reset it to zero. Make a note of the cumulative distance moved by the dye up to this point. Reset the respirometer by pushing the plunger to move the coloured dye to the start position.

As soon as the dye starts moving again, restart the stopwatch and continue recording the cumulative distance moved by the coloured dye at the completion of each 30-second interval, up to 5 minutes. These measurements will need to include both the cumulative distance noted when the respirometer was reset and the distance moved subsequently.

(a) Prepare a table and record your results below.

[3]

- (b)** Suggest why the mung beans were left in the respirometer for 5 minutes before pulling in the dye.

.....
..... [1]

- (c)** Suggest an advantage of recording the cumulative distance moved by the coloured dye every 30 seconds, instead of only recording the total distance moved after 5 minutes.

.....
..... [1]

- (d)** Explain why the coloured dye move along the capillary tube.

.....
.....
.....
..... [2]

The experiment can be conducted three times to obtain a mean reading.

- (e)** Identify one other method that can improve the accuracy of the experiment.

.....
..... [1]

[Total: 8]

Question 3

Slide **R** is a stained transverse section through a plant root.

You are not expected to be familiar with this specimen.

(a) Draw a large plan diagram for the whole root.

You are expected to draw the correct shape and proportions of the different tissues.

Use a sharp pencil for drawings.

[3]

(b)

- (i)** Observe the cells in the cortex region of the roots on slide **R**.

Select four adjacent, touching cells which show observable features of the cortex tissue. Each cell must touch at least two other cells.

Make a large drawing of this group of four touching cells.

Use ruled label line to identify the cell wall of one cell.

[4]

- (ii)** Using the stage micrometer, measure the actual length of one of the cell drawn in **(b)(i)**.

Show all the steps in your calculation, including the appropriate units.

Length of cell =
[2]

- (iii)** Calculate the magnification of the drawing in **(b)(i)**.
Show all the steps in your calculation, including the appropriate units.

Magnification =
[2]

(c) Fig. 3.1 is a photomicrograph of a stained transverse section through a root of a different type of plant.

You are not expected to be familiar with this specimen.

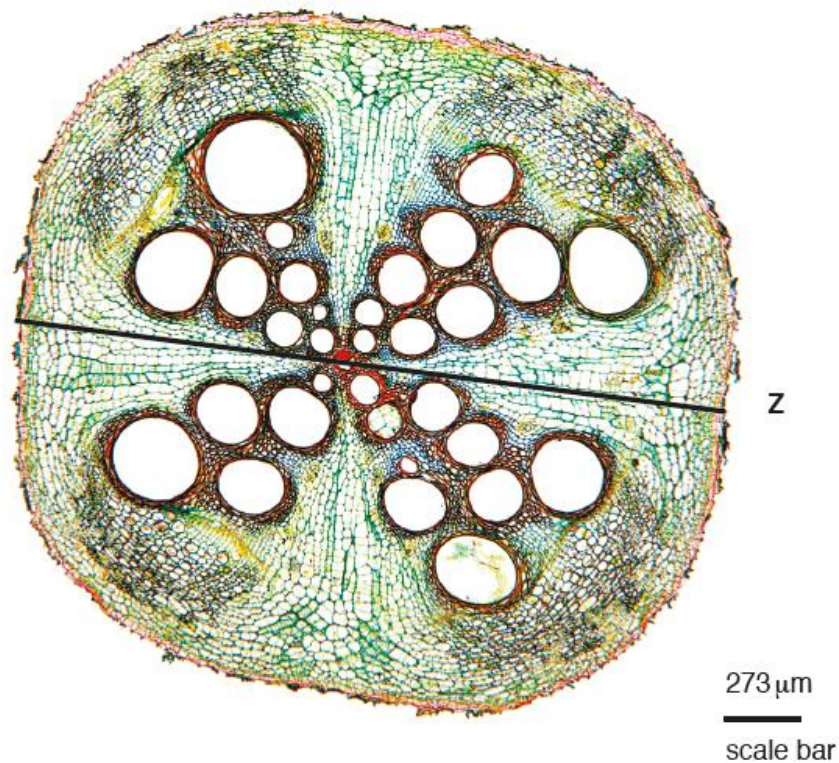


Fig. 3.1

Using the scale bar, measure the length of diameter of the root in Fig. 3.1, denoted by line Z.

Length of root in Fig. 3.1
[3]

[Total: 14]

End of Paper